MIL-P-46125A(MR)
11 April 1980
SUPERSEDES
MIL-P-46125(MR)
21 September 1967

MILITARY SPECIFICATION

PROJECTILE CUBES, FRAGMENT-SIMULATING

This specification is approved for use by the Army Materials and Mechanics Research Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers fragment-simulating projectile cubes used in ballistic testing (see 6.1).

1.2 Classification. The fragment simulating projectile cubes shall be classified as follows:

Nominal weight (grains)
2
4
16
64
100
225

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

STANDARDS

FEDERAL

Fed. Test Method Std. No. 151 - Metals; Test Methods

/FSC 1305/

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Army Materials and Mechanics Research Center, Watertown, MA 02172 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ANSI B46.1 Surface Texture

(Application for copies should be addressed to American National Standards Institute, 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 331 - Steel Bars, Alloy, Cold-Finished
A 751 - Standard Methods, Practice and Definitions for Chemical Analysis of Steel Products
E 18 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Composition. The fragment-simulating projectiles shall be manufactured from cold rolled, annealed, steel conforming to composition of ASTM A 331.

3.2 Hardness. The fragment-simulating projectiles shall be fully quenched and tempered to a hardness value of Rockwell C29 $\pm$ 2 throughout.

3.3 Dimensions and tolerances. Opposite surfaces of fragment simulating projectile cubes shall be parallel. Dimensions and tolerances of fragment-simulating projectiles shall be as shown in table I.

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Table I. Dimensions and tolerances

Nominal weight (grains)	Dimensions (inch cube)	Tolerance (inch)
2	0.100	± 0.001
4	0.126	± 0.001
16	0.200	± 0.001
64	0.317	± 0.001
100	0.370	± 0.001
225	0.486	± 0.001

3.4 Finish. The surface finish of the fragment-simulating projects shall be 63.

3.5 Workmanship. The fragment simulating projectiles shall have sharp edges and corners and shall be free from burrs and flakes.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Lot. A lot shall consist of all fragment-simulating projectiles of the same size.

4.3 Sampling.

4.3.1 For chemical composition. At least one sample shall be selected for chemical analysis from the bars used for manufacturing the fragment-simulating projectiles prior to machining or from sufficient parts prior to heat treating.

4.3.2 For hardness tests. Sampling shall be in accordance with general inspection level II of MIL-STD-105. The acceptable quality level (AQL) shall be 2.5 percent defective.

4.3.3 For dimensions, finish and examination. Sampling shall be in accordance with general inspection level II of MIL-STD-105. The acceptable quality level (AQL) shall be 2.5 percent defective.

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4.4 Examination.

4.4.1 Visual. Each sample shall be visually inspected to determine compliance with the workmanship requirements.

4.4.2 Dimensions and finish. Measurements shall be made on each sample to determine compliance with the requirements dimensions and finish.

4.5 Tests.

4.5.1 Chemical analysis. Chemical analysis for compliance with the material composition requirement of 3.1 shall be in accordance with ASTM A 751.

4.5.2 Hardness test. Rockwell C-scale hardness (Rc) readings shall be taken on the surface of each sample in accordance with ASTM E 18. When it is impractical to take Rc readings because of size limitations, Rockwell Superficial hardness readings may be taken in accordance with ASTM E 18.

4.6 Retest and rejection. Retest and rejects shall be in accordance with the general section of Fed. Test Method Std. No. 151.

5. PACKAGING

5.1 Preservation, packaging, and packing.

5.1.1 Commercial. Unless otherwise specified (see 6.2) fragment-simulating projectiles shall be preserved, packaging and packed in accordance with MIL-STD-1188.

5.2 Marking for shipment. In addition to any special marking specified in the contract or order, marking for shipment shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The fragment simulating projectiles covered by this specification are intended for use in ballistic testing of various materials.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Nominal weight and dimensions (see 3.3)

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Project No. 1305-A899

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